Economics S5040: Dynamic Stochastic Modeling
Lecture 20 — Recursive Asset Pricing in a Markov Economy

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Central question: How do we price an asset that pays dividends repeatedly and can itself be resold next period? The answer is a recursive pricing equation that closely resembles a Bellman equation.

Plan for an 80-minute class

Minutes	Topic
0–10	Review: Q as a one-period pricing operator
10–27	Deriving the recursive pricing equation
27–42	Matrix solution and present-value interpretation
42–55	Price-dividend ratios and state dependence
55–68	Multi-period bonds and the term structure
68–76	Expected returns and the market portfolio
76–80	Bellman analogy and summary

1 Review: the one-period pricing operator

From Lecture 19,

$$m_{ij} = \beta \frac{u'(c_j)}{u'(c_i)}, \qquad q_{ij} = P_{ij} m_{ij}.$$

Let

$$Q = [q_{ij}].$$

For any one-period payoff vector x,

$$\boxed{p = Qx.}$$

Thus Q is a linear operator that maps next-period state-contingent consumption into current prices.

2 A long-lived dividend claim

Now consider an asset that pays a dividend every period and can be sold next period. Let

$$d_j$$

be next period's dividend if state j occurs, and let

$$v_j$$

be the asset's ex-dividend price next period in state j.

If today's state is i, buying the asset gives total next-period payoff

$$d_j + v_j.$$

Therefore

$$v_i = \sum_j P_{ij} m_{ij} (d_j + v_j).$$

In matrix notation,

$$\boxed{v = Q(d + v).}$$

Recursive pricing equation. Today's price equals the discounted expected value of next period's dividend plus next period's resale price.

3 Solving the recursive pricing equation

Starting from

$$v = Qd + Qv,$$

move Qv to the left:

$$(I - Q)v = Qd.$$

If $I - Q$ is invertible,

$$\boxed{v = (I - Q)^{-1} Qd.}$$

This formula looks compact, but it contains an infinite sequence of future dividends.

If the spectral radius satisfies

$$\rho(Q) < 1,$$

then

$$(I - Q)^{-1} = I + Q + Q^2 + Q^3 + \cdots .$$

Hence

$$\begin{aligned} v &= (I - Q)^{-1} Qd \\ &= (I + Q + Q^2 + \cdots) Qd \\ &= Qd + Q^2 d + Q^3 d + \cdots . \end{aligned}$$

Therefore

$$\boxed{v = \sum_{n=1}^{\infty} Q^n d.}$$

The term $Q^n d$ is the current value of dividends paid n periods from now.

4 Why powers of Q appear

Consider a two-period payoff. To value consumption delivered two periods ahead, we must price it across every possible path

$$i \to j \to k.$$

The one-period state price along the first step is q_{ij} and along the second step is q_{jk}. Summing over intermediate states produces matrix multiplication:

$$(Q^2)_{ik} = \sum_j q_{ij} q_{jk}.$$

Thus Q^2 prices two-period state-contingent payoffs, Q^3 prices three-period payoffs, and so on.

This is the same reason powers of a Markov transition matrix describe multi-period transition probabilities. The difference is that Q incorporates both probabilities and marginal-utility discounting.

5 A numerical finite-state economy

Continue with the three-state economy from Lecture 19:

$$c = \begin{pmatrix} 0.95 \\ 1.00 \\ 1.05 \end{pmatrix}, \qquad P = \begin{pmatrix} 0.80 & 0.15 & 0.05 \\ 0.10 & 0.80 & 0.10 \\ 0.05 & 0.15 & 0.80 \end{pmatrix},$$

with

$$\beta = 0.96, \qquad \rho = 2.$$

Suppose dividends are positively related to aggregate consumption:

$$d = \begin{pmatrix} 0.80 \\ 1.00 \\ 1.25 \end{pmatrix}.$$

The computational steps are:

1. Construct $M = [m_{ij}]$.
2. Construct $Q = P \circ M$.
3. Compute

$$v = (I - Q)^{-1} Q d.$$

4. Report v_i and the price-dividend ratio v_i/d_i in each state.

Before calculating, ask:

> Should the price-dividend ratio necessarily be the same in low-, normal-, and high-consumption states?

No. Both expected future dividends and the discounting of those dividends depend on the current state.

6 Price-dividend ratios

Define

$$\frac{v_i}{d_i}$$

as the price-dividend ratio in state i.

A high price-dividend ratio can arise because future dividends are expected to be high, because future payoffs are discounted less heavily, or both. In a Markov economy, today's state changes the conditional distribution of tomorrow's state, so valuation ratios naturally become state dependent.

This is an important conceptual departure from a constant-discount-rate present-value formula.

7 Multi-period zero-coupon bonds

The pricing operator also gives a simple recursion for bonds.

Let $b_i^{(n)}$ denote the price in current state i of a zero-coupon bond that pays one unit of consumption in every state n periods from now. Define

$$b^{(0)} = \mathbf{1}.$$

Then

$$\boxed{b^{(n+1)} = Qb^{(n)}.}$$

Therefore

$$\boxed{b^{(n)} = Q^n\mathbf{1}.}$$

The continuously compounded n-period yield in state i is

$$\boxed{y_i(n) = -\frac{1}{n}\log b_i^{(n)}.}$$

Thus the model generates an entire state-dependent yield curve from the same pricing matrix used to price risky assets.

Unifying idea. Once we know Q, we can price one-period claims, long-lived dividend claims, and bonds of many maturities.

8 The market portfolio

In a representative-agent endowment economy, aggregate consumption is the economy's aggregate dividend. A claim to all future aggregate consumption is therefore the natural market portfolio.

Set

$$d = c.$$

Then

$$\boxed{v = (I - Q)^{-1} Qc}$$

is the price vector for the claim to future aggregate consumption.

If the current state is i and next period's realized state is j, the gross market return is

$$\boxed{R^m_{ij} = \frac{c_j + v_j}{v_i}.}$$

The expected market return conditional on state i is

$$\mathbb{E}_i[R^m] = \sum_j P_{ij} R^m_{ij}.$$

Compare this with

$$R^f_i = \frac{1}{\sum_j q_{ij}}.$$

The conditional risk premium is

$$\mathbb{E}_i[R^m] - R^f_i.$$

The market portfolio is risky because its payoff is tied to aggregate consumption. It tends to pay less in states in which marginal utility is high, so investors require compensation for bearing that aggregate risk.

9 Recursive pricing versus a Bellman equation

There is a useful structural analogy with dynamic programming.

A Bellman equation has the form

$$V(s) = \max_a \left\{ r(s,a) + \beta \mathbb{E}[V(s') \mid s, a] \right\}.$$

The asset-pricing recursion has the form

$$v(s) = \mathbb{E}\left[m(s,s') \{ d(s') + v(s') \} \mid s \right].$$

Both equations are recursive: the current object depends on a current or next-period payoff plus the continuation value of the same object.

But there is an important difference:

- In dynamic programming, the agent chooses a control and maximizes.

- In asset pricing, the optimal consumption allocation has already generated the SDF. Given m, pricing is a linear recursion with no maximization step.

Connection to the course. Dynamic programming taught us how optimal behavior generates continuation values. Asset pricing uses marginal utility from that optimal behavior to value claims to future consumption.

10 Long-run pricing and the dominant eigenvalue

Because long-maturity prices involve large powers of Q, eigenvalues matter. If $\lambda_{\max}$ is the dominant eigenvalue of Q, then for large n, under standard regularity conditions,

$$Q^n \approx \lambda_{\max}^n w r',$$

where w and r are corresponding right and left eigenvectors.

Thus long-maturity bond prices decay approximately at rate $\lambda_{\max}^n$. A corresponding asymptotic continuously compounded yield is

$$\boxed{y_\infty = -\log(\lambda_{\max}).}$$

This gives a direct connection between the linear algebra studied earlier in the course and long-run asset pricing.

11 A computational roadmap

A Jupyter notebook for this lecture should proceed in the following order:

1. Enter c, P, β, and ρ.
2. Compute M using
$$m_{ij} = \beta(c_j/c_i)^{-\rho}.$$
3. Compute $Q = P \circ M$.
4. Compute one-period risk-free rates.
5. Price a one-period risky payoff x using $p = Qx$.
6. Price the perpetual dividend claim using
$$v = (I - Q)^{-1} Q d.$$
7. Compute price-dividend ratios.
8. Compute $b^{(n)} = Q^n \mathbf{1}$ and graph yields for several maturities.
9. Set $d = c$ and compute state-contingent market returns and conditional expected returns.

Summary

1. The pricing matrix Q is a one-period pricing operator.
2. A long-lived asset satisfies
$$v = Q(d + v).$$
3. Therefore
$$v = (I - Q)^{-1}Qd = \sum_{n=1}^{\infty} Q^n d.$$
4. Multi-period bond prices satisfy
$$b^{(n)} = Q^n \mathbf{1}.$$
5. Price-dividend ratios, risk-free rates, expected returns, and yield curves can all depend on the current Markov state.
6. Large powers of Q connect long-run asset pricing to its dominant eigenvalue.

The Week 10 chain:

Markov consumption $\to$ marginal utility $\to M \to Q \to$ asset prices and returns.